

MAJOR PROJECT REGISTRATION FORM

Date: 31-07-2026

Project Category:	☒ Research oriented	☐ Application oriented	Batch ID:	CSM-C15
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: DeepRealGuard: Multi-Modal Deepfake Detection with Explainable AI

Abstract

The rapid advancement of Artificial Intelligence (AI) and Generative AI has enabled the creation of highly realistic deepfake videos, posing significant threats to digital trust, cybersecurity, journalism, law enforcement, finance, and social media platforms. Deepfake content can spread misinformation, manipulate public opinion, damage reputations, facilitate identity theft, and compromise the authenticity of digital evidence. Existing deepfake detection systems primarily rely on single-modal visual analysis, making them less effective against compressed, edited, low-quality, and unseen deepfake videos. Furthermore, most existing approaches function as black-box models, providing limited explainability and lacking mechanisms to verify the authenticity and origin of digital content.

The proposed project, **“DeepRealGuard – Multi-Modal Deepfake Detection with Explainable AI”** addresses these challenges by developing a robust, transparent, and secure deepfake detection framework. Inspired by the IEEE research paper "Towards Robust Deepfake Detection in the Wild: A Multi-Modal and Explainable Approach" (2025), the proposed system extends existing research by integrating multimodal analysis, Explainable AI, and blockchain-based provenance verification into a single platform.

The system combines three complementary sources of information: visual features extracted from video frames, audio spectrogram analysis, and multimedia metadata. Video frames are analyzed using a hybrid 3D Convolutional Neural Network (3D-CNN) and Vision Transformer (ViT) architecture, audio features are processed using Convolutional Neural Networks (CNNs), and metadata is classified using XGBoost. These features are fused through a Cross-Attention Fusion mechanism to improve detection accuracy and robustness. To enhance transparency, Grad-CAM and Layer-wise Relevance Propagation (LRP) generate visual heatmaps highlighting manipulated regions. The system also incorporates adversarial training to improve resistance against compression, cropping, resizing, and other post-processing attacks.

The model is trained and evaluated using benchmark datasets, including FF++ (2025 Update), Celeb-DF v3 (2025), Isolated Voices (2025), and the Social Media Deepfakes Dataset (2025). The application is developed using React.js, FastAPI, PyTorch, MongoDB, Docker, and AWS Cloud. The expected outcome is a scalable, highly accurate, and explainable deepfake detection platform that strengthens digital content authentication, reduces misinformation, supports forensic investigations, and enhances user trust through transparent AI predictions and secure provenance verification.

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References (Category specific)

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